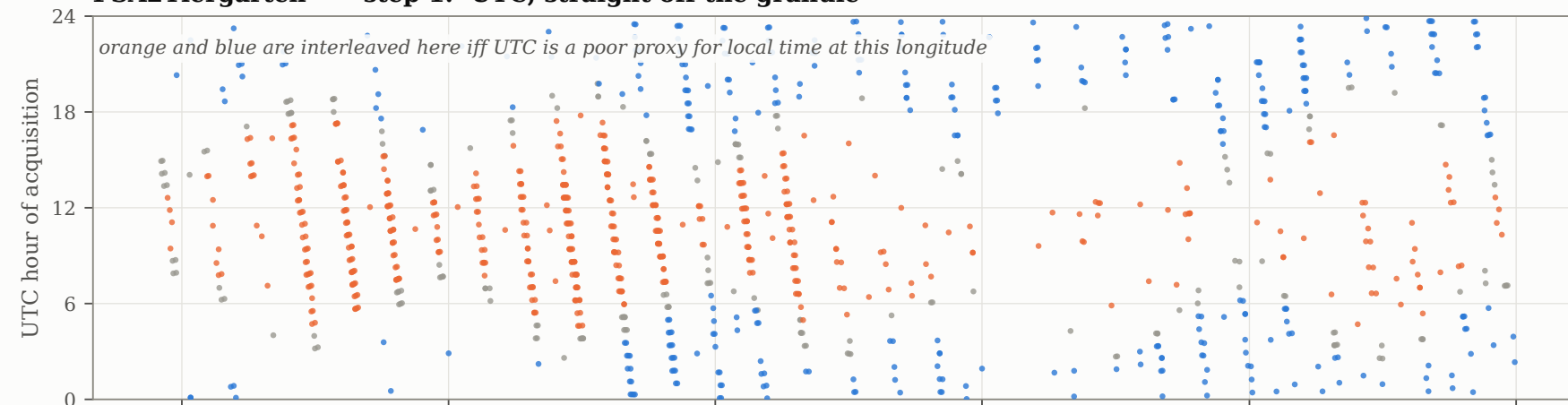
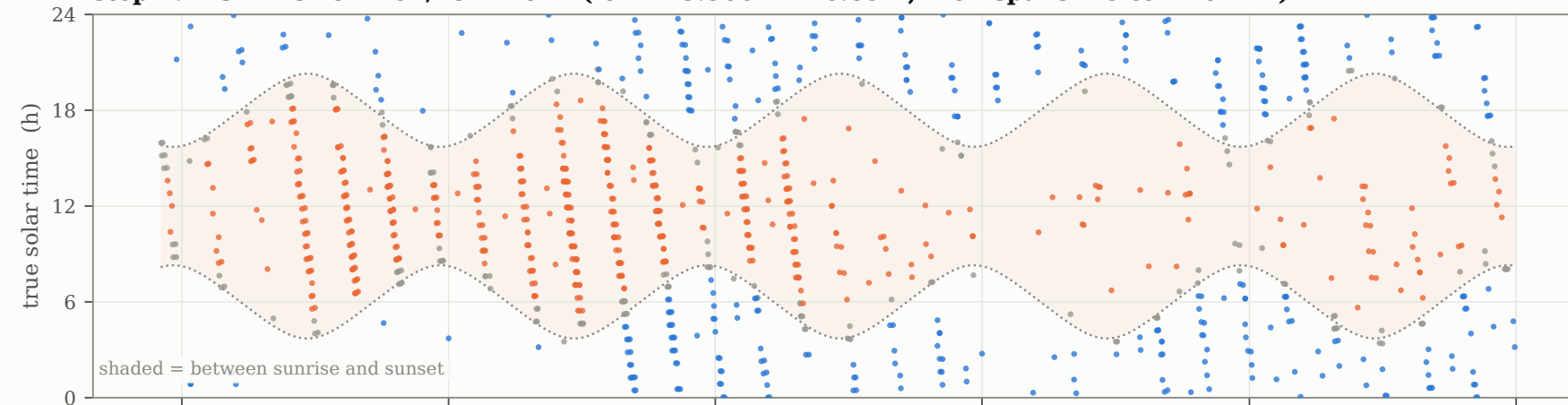


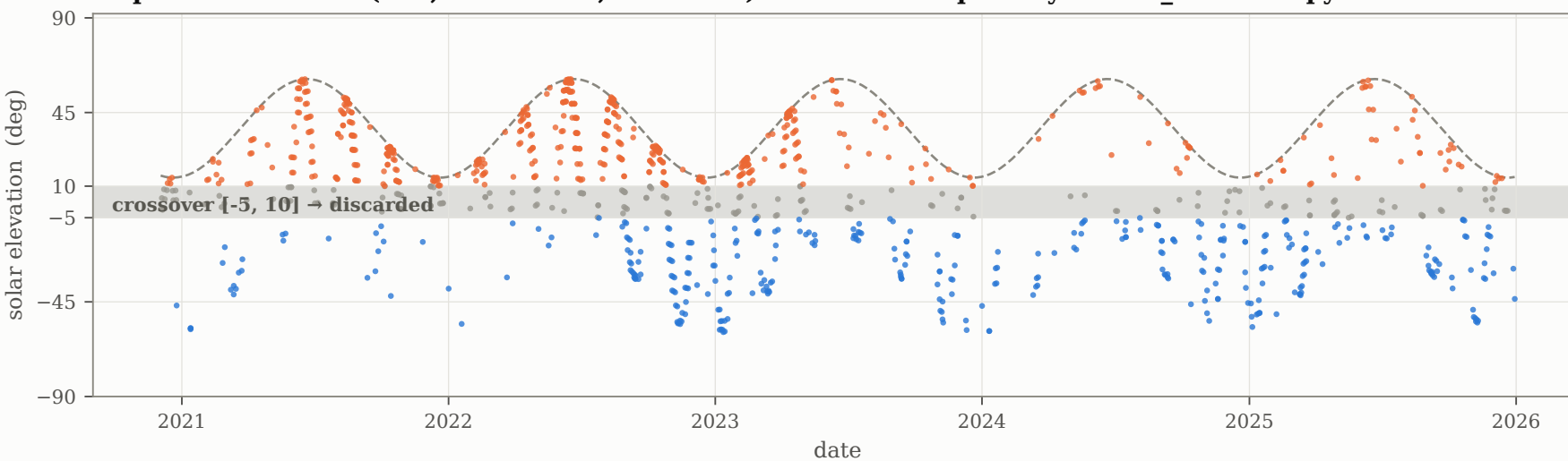
PSA2Tiergarten — step 1: UTC, straight off the granule



step 2: TST = UTC + lon/15 + EoT (lon +13.366° → +0.89 h; EoT spans -15 to +16 min)



step 3: elevation = f(TST, declination, lat 52.51°) — the ONLY quantity census_ecostress.py:241 classifies on



worked examples

UTC	TST	elev	phase
2020-12-12T12:37	13.60	11.4	day
2021-08-01T17:16	18.07	13.5	day
2021-10-20T08:15	9.42	18.0	day
2022-05-21T21:26	22.39	-14.4	night
2022-08-07T14:05	14.89	40.3	day
2022-10-17T10:33	11.70	27.0	day
2023-02-02T13:32	14.20	14.6	day
2023-04-21T08:13	9.13	36.5	day
2023-11-24T18:08	19.23	-27.7	night
2024-10-30T02:44	3.91	-30.8	night
2025-05-14T21:06	22.06	-14.6	night
2025-12-30T02:19	3.17	-43.6	night

longitude offset +0.89 h
median |TST - UTC| 0.90 h

UTC cannot be read as local time here: the same UTC hour lands in daylight or in darkness depending on the date.